Strategy Report: What Is Linked to Engagement on Trending YouTube Videos?

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What this project was about

I wanted to understand why some YouTube videos get far more likes and comments than others. To explore this, I looked at 2,143 trending US videos and called the top 20% by engagement rate "viral" for this report. A Random Forest model picked up a clear pattern, and it did a good job of ranking high-engagement videos above ordinary ones (AUC 0.75). I see it as a helpful guide to what goes with engagement, not a fixed formula.

Three things stood out for me:

- Videos with warmer, more positive comment sections tended to have higher engagement.
- Most of the useful signal came from things a creator can choose, like the title and the category.
- Shorter titles went a little better than long ones.

The data I used

I used the public Kaggle dataset *Trending YouTube Video Statistics* (`datasnaek/youtube`), which has video statistics (`USvideos.csv`) and comments (`UScomments.csv`). The course support team confirmed that using Kaggle data is allowed. Every video in the dataset reached the trending page at some point.

What I did

I measured **engagement rate** as likes plus comments divided by views, using each video's last day on the trending page. Videos in the top 20% were marked "viral".

For each video I looked at:

- **Relatability ratio:** the share of its comments that VADER (a sentiment tool) scored as strongly positive. I only used videos with at least 20 scored comments, so the ratio would be based on enough comments.
- **Hype velocity:** the views a video already had on its first day on the trending page.
- **Title features:** character length, word count, and whether it had a question mark or exclamation mark.

- **Category:** the YouTube category ID.

I trained a Decision Tree and a Random Forest on 80% of the videos and tested them on the remaining 20%. I also repeated the test on 5 different splits to check the result held up, and used permutation importance to see which features the model relied on most.

How the model did

Measure	Result
Videos analysed	2,143
Videos in the "viral" group	about 20%
Accuracy	70.4% (above the 51% requirement)
AUC	0.75 on the test set, 0.76 across 5 splits
Viral videos the model found (recall)	55%
Correct when it says "viral" (precision)	35%

Since only about 1 in 5 videos is in the viral group, accuracy alone doesn't tell the whole story, so I also looked at AUC and recall. The model found more than half of the high-engagement videos, and the AUC stayed steady across all 5 splits, which makes me comfortable that the pattern is real.

What I found

1. The mood of the comments mattered most. The relatability ratio came first, followed by category and then title length. I had guessed that fast early growth would lead, so this surprised me. Positive comments and high engagement probably both come from viewers who already enjoy a video, so I see this as a link rather than a cause.

2. Shorter titles did a little better. The correlation was -0.23, which is a gentle effect, so I treat it as a tendency and not a rule. Exclamation marks and question marks added very little.

3. Category and title carry a lot on their own. Using only title and category, the model reached an AUC of 0.72, compared with 0.76 using everything. In the dashboard, Music and How-to & Style had high viral shares (35.6% and 33.6%, compared with 20% overall). So the niche seems to give a video a good starting point, though it doesn't guarantee anything.

What creators can take from this

These are ideas worth trying, not promises.

1. **Try shorter titles.** Trimming filler words and keeping titles from being cut off on phones looks like a good habit. A simple A/B test on two titles would show whether it works for a specific channel.
2. **Pick the category with care.** It was one of the strongest signals, so choosing the right niche for the audience seems worthwhile.
3. **Treat comments as friendly feedback.** A warm comment section is a good sign that a video is connecting with people, and it can guide what to make next.
4. **Spend time on the words, not the punctuation.** Exclamation and question marks didn't change much, so the wording and length of the title are a better use of effort.

A few things to keep in mind

- The results show patterns, not proof of what causes engagement.
- Only trending videos are included, so the findings describe videos that were already doing well on YouTube.
- Comments come in after a video is published, so they are a sign of how people reacted and not something a creator can set up in advance.
- The model is a helpful guide: it finds a bit over half of the high-engagement videos, and about 1 in 3 of the videos it points to turn out to be in that group.

Where this could go next

It would be great to look at a wider set of videos, not only trending ones, and to run real A/B tests on titles. That would help turn these patterns into tested advice for creators.